

Group Theory

Introduction

1.

- (a) There are two different isomorphism classes of groups of order 12 which can be written as $\mathbb{Z}/6\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$. What are they?
- (b) Suppose G is a group with a normal subgroup N , and let $\pi: G \rightarrow G/N$ be the projection to the quotient. Prove that G is a(n internal) semi-direct product $N \rtimes G/N$ if and only if there exists a homomorphism $s: G/N \rightarrow G$ such that $\pi \circ s = \text{id} : G/N \rightarrow G/N$. That is to say, if and only if the exact sequence of groups

$$1 \rightarrow N \rightarrow G \rightarrow G/N \rightarrow 1$$

splits on the right.

2. (August 2018 Problem 2) For a finite group G , denote by $s(G)$ the number of subgroups of G .

- (a) Show that $s(G)$ is finite.
- (b) Show that if H is a nontrivial normal subgroup of G , then $s(G/H) < s(G)$.
- (c) Show that $s(G) = 2$ if and only if G is cyclic of prime order.
- (d) Show that $s(G) = 3$ if and only if G is cyclic of order p^2 for a prime p .

3. (January 2018 Problem 4) Let G be a finite group. Denote by $\text{Aut}(G)$ the group of automorphisms of G , and by $Z(G) \subset G$ the center of G .

- (a) (Slightly modified) Let $\text{Inn}(G) \subset \text{Aut}(G)$ be the subgroup consisting of automorphisms coming from conjugation, i.e. automorphisms of the form $x \mapsto gxg^{-1}$ for some $g \in G$. Prove that $\text{Inn}(G)$ is isomorphic to $G/Z(G)$.
- (b) Show that if $G/Z(G)$ is cyclic, then G is abelian.
- (c) Show that if $\text{Aut}(G)$ is cyclic, then G is abelian.
- (d) Show that if G is abelian, then $x \mapsto x^{-1}$ is an automorphism of G . (In fact this is an if and only if.)
- (e) Deduce that there is no group G such that $\text{Aut}(G)$ is a nontrivial cyclic group of odd order.

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4. (January 2017 Problem 1) Give an example of each of the following.

- (a) A group G with a normal subgroup N such that G is not a semidirect product $N \rtimes G/N$.
- (b) A finite group G that is nilpotent but not abelian.
- (c) A group G whose commutator subgroup $[G, G]$ is equal to G .
- (d) A non-cyclic group G such that every Sylow subgroup of G is cyclic.
- (e) A transitive action of S_3 on a set X of cardinality greater than 3.

5. Prove that any nilpotent group is solvable.

6. Give an example of a group G with subgroups $N \supseteq H$ so that:

- N is normal in G ,
- H is normal in N ,
- H is *not* normal in G .

7. (August 2021 Problem 3) Let G be a finite group. For given $n > 1$, we say that G admits roots of degree n if, for every $x \in G$, there exists $y \in G$ such that $y^n = x$.

- (a) Show that if n is coprime to $|G|$, G admits roots of degree n .
- (b) Conversely, prove that if G admits roots of degree n , then n is coprime to $|G|$.

8. (August 2019 Problem 5) Let G be the group of 2×2 invertible upper triangular matrices over the field $\mathbb{F}_p$.

- (a) Prove that G has only one subgroup of order p , and that this subgroup is normal. (Hint: first show that an element of order p must have 1's on the diagonal.)
- (b) Prove that G is solvable by exhibiting a homomorphism f from G to an abelian group A such that the kernel of f is also abelian.
- (c) Prove that if $p \neq 2$, G is not nilpotent.

9. (January 2020 Problem 5)

- (a) Find a nontrivial finite subgroup of $\mathrm{GL}_2(\mathbb{R})$.
- (b) Find an infinite, abelian, but non-cyclic subgroup of $\mathrm{GL}_2(\mathbb{R})$.
- (c) Find a proper, normal subgroup of $\mathrm{GL}_2(\mathbb{Z})$ of finite index.